Risk-Adjusted Compounding with the Convex Core Portfolio

A plain-English guide for do-it-yourself ETF and mutual-fund investors - how Growth, Duration, and Convexity work together to compound wealth while reducing deep drawdown risk.

> This is the everyday-investor version of a longer research note. No math is required to use it. Convex Core is a simplified, practical synthesis of several of the most durable ideas in risk-adjusted compounding: geometric-growth logic, broad diversification, risk budgeting, trend-following crisis alpha, and optional volatility management.

The one idea: don't dig deep holes

Most investing advice chases higher returns. This portfolio is built around a quieter truth: **avoiding big losses matters more than catching big gains**, because losses are asymmetric. Lose 50% and you don't need +50% to get back – you need +100%. The deeper the hole, the more punishing the climb out.

LONG-RUN GROWTH, IN ONE LINE

$$g \approx \mu - \tfrac{1}{2}\sigma^2$$

Your long-term growth (g) is roughly your average return minus a penalty for bumpiness.

In plain English: **growth of investment assets = average returns - a volatility/drawdown penalty**

The penalty is subtracted, so smoothing the ride - especially cutting deep drops - can raise real compounded growth.

PORTFOLIO ALLOCATION, IN ONE LINE

Wgrowth + Wconvex + Wduration = 1

Wgrowth = growth sleeve weight Wconvex = convexity sleeve weight Wduration = duration sleeve weight

Three jobs, three sleeves

Instead of guessing what will go up, the portfolio assigns three jobs and fills each with the right kind of fund - one engine and two different kinds of insurance.

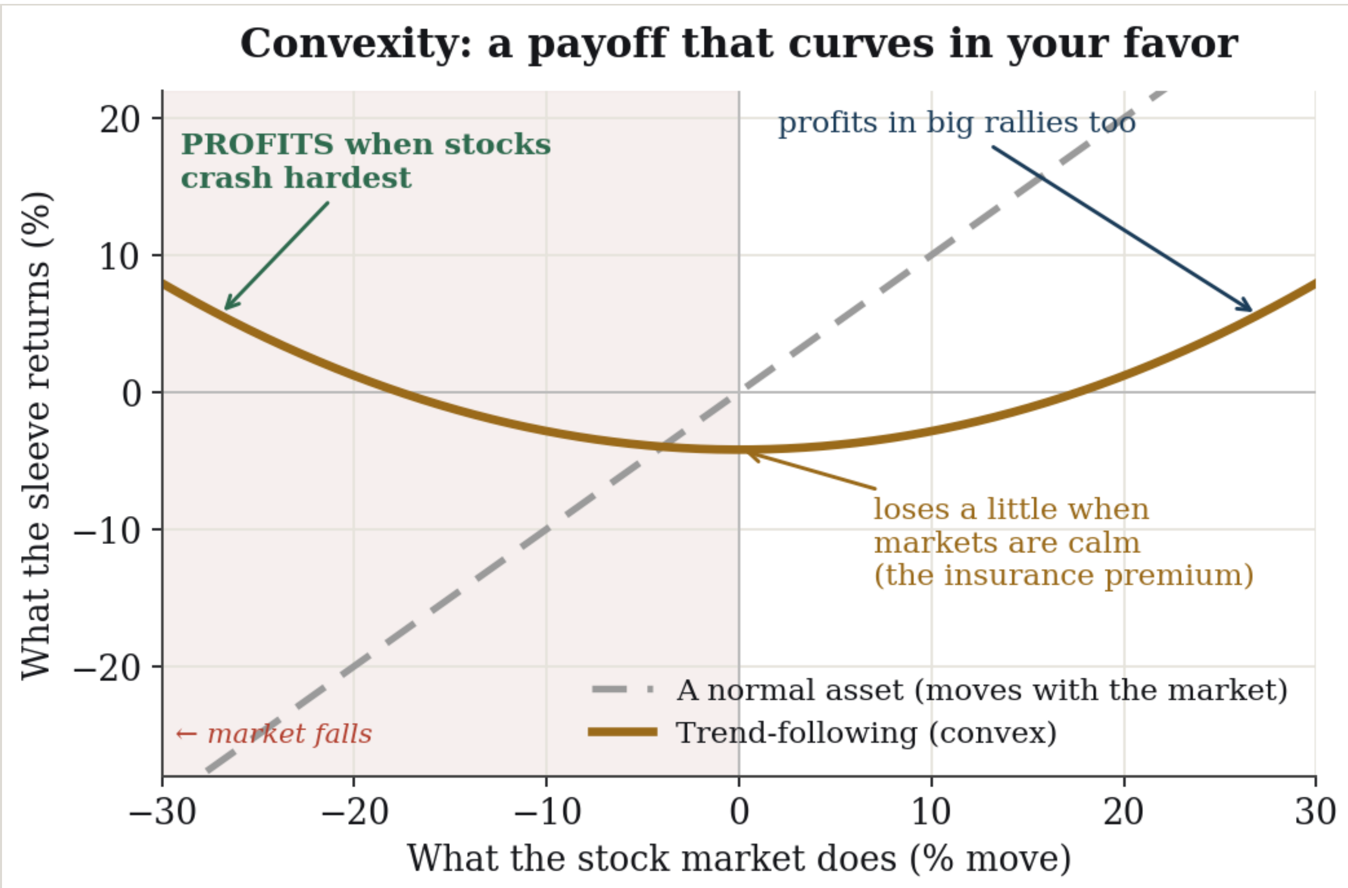
GROWTH Plain stock ownership - the long-run source of returns. The biggest slice.

DURATION High-quality bonds (Treasuries). In a recession/deflation panic - 2008, 2020 - money can flood into Treasuries while stocks fall.

CONVEXITY Trend-following managed futures. In inflation or rate shocks, this sleeve can help when stocks and bonds are both under pressure.

What "convexity" actually means

Convexity is just a payoff that curves in your favor. As the market moves more, a convex holding's gains grow faster and its losses shrink — the payoff line bends upward instead of running straight. The opposite, concavity, bends against you (small steady gains, then a blow-up in a crisis). A convex position likes big moves in either direction; in the trade it's called being "long volatility."



A normal asset (grey) moves one-for-one with the market. A convex holding (gold) bends upward at both ends — it makes money in big moves either way and gives a little back when markets are calm.

Read the gold curve: when stocks fall hard (left side), it rises — that's the part that protects you. When markets are quiet (the middle), it loses a little — that small, steady cost is the insurance premium you pay for the protection. This is exactly the shape you want bolted onto a stock portfolio, because the protection shows up precisely when your stocks are getting hurt.

What are "managed futures" and "trend-following"?

Managed futures — funds that buy and sell futures contracts (standardized agreements to buy or sell stocks, bonds, currencies, or commodities at a set future price). The key feature: they can go both **long** (bet a price rises) and **short** (bet a price falls). That ability to profit from falling markets is what lets them help in a crash.

Trend-following — the most common managed-futures strategy, and a purely rules-based one: **buy what's been going up, short what's been going down, and ride the move until it reverses.** No forecasting, no opinions — it simply reacts to price.

How that produces convexity

Put those two together and the convex smile falls out naturally:

- **In a sustained crash,** prices trend down. The fund flips short early and earns more the further markets fall — so the payoff bends up on the left (the crash side).
- **In a sustained boom,** it's long and rides the move up — so it gains on the right too.
- **In choppy, directionless markets** (the middle), trends keep starting and reversing, and the fund gets whipsawed into small losses — the dip in the smile, i.e. the premium.

Researchers Fung & Hsieh showed this payoff statistically resembles owning a straddle — holding both a put and a call, which profits whenever the price moves a lot in either direction. That's the textbook definition of a convex, long-volatility position. Trend-following gets you that shape without buying options.

An example portfolio you could actually build

Illustrative weights of **65% growth / 15% convexity / 20% duration** — the research model's split. Funds are examples, not recommendations; swap equivalents freely (e.g., SPY for VOO).

SLEEVE	WHAT IT DOES	EXAMPLE FUNDS	WEIGHT
GROWTH	US large-cap core	VOO or SPY	38%
	International stocks	VXUS	15%
	US small-cap value tilt	AVUV	12%
Growth subtotal			**65%**
CONVEXITY	Managed-futures ETF (replication)	DBMF	8%
	Trend-following mutual fund	AQMNX / AQMIX	7%
Convexity subtotal			**15%**
DURATION	Treasuries (tax-free accounts)	IEF / VGIT / TLT / GOVT	20%
	— or in a taxable account — municipal bonds	VTEB / MUB	(20%)
Duration subtotal			**20%**

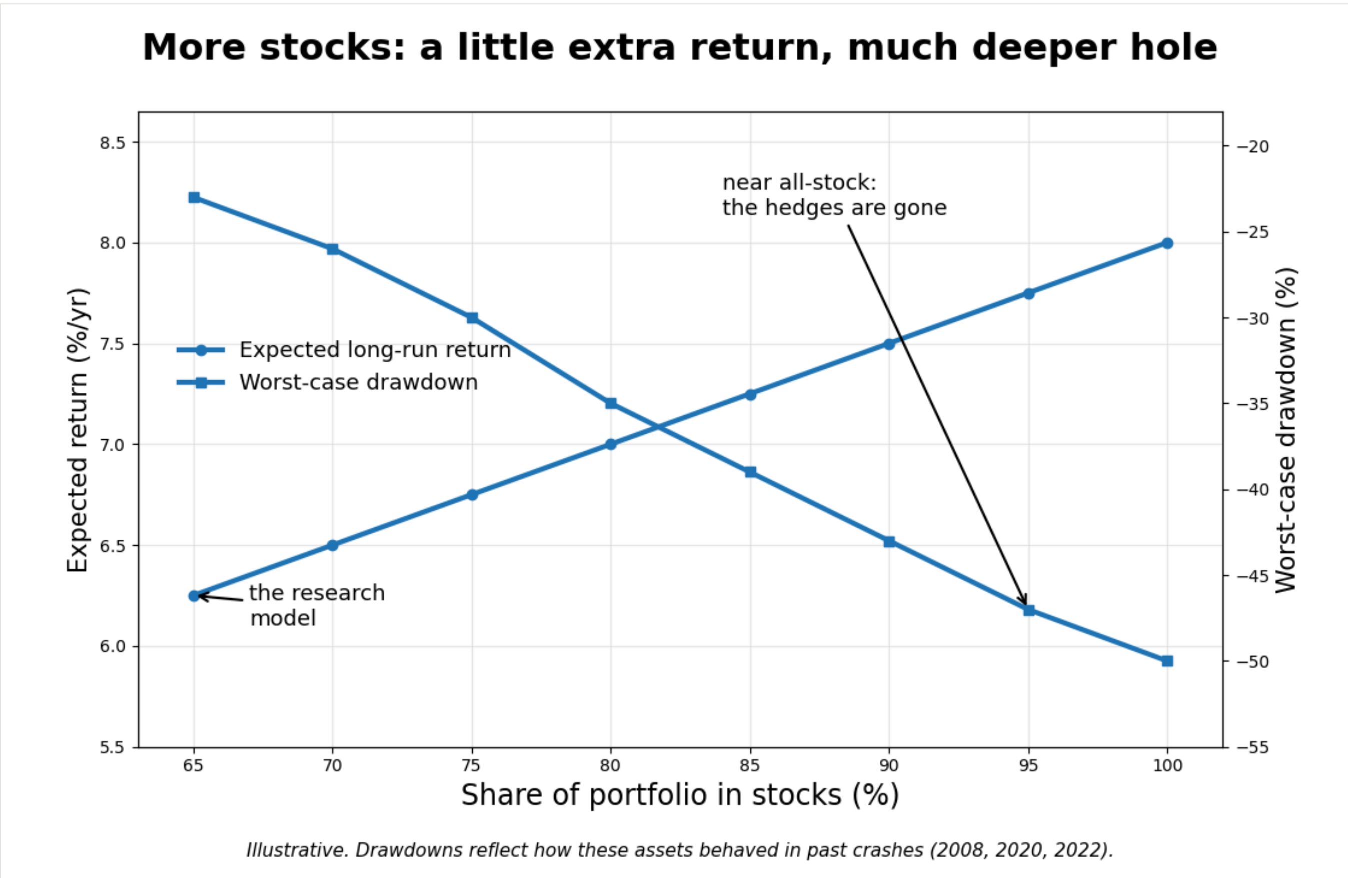
Why two funds in the convexity sleeve?

DBMF (iMGP DBi Managed Futures Strategy ETF) and the **AQR Managed Futures** fund (AQMNX retail, or cheaper AQMIX) do the same job differently, so holding both is sensible diversification, not redundancy:

- **DBMF** is an ETF that replicates the average big managed-futures fund — lower-cost, easy to buy anywhere, and (as an ETF) relatively tax-friendly.
- **AQR Managed Futures** is a hands-on trend-following mutual fund with a long record and a slightly negative tie to stocks. It trades a lot, so it's pricier and tax-inefficient — which is why it belongs in a retirement account.

"What if I just hold more stocks?"

A fair question. Stocks have the highest long-run return, so why not push the equity slice from 65% up toward 95% or 100%? You can — but be clear about the trade you're making. More stocks buys you a little extra expected return and a lot more downside.

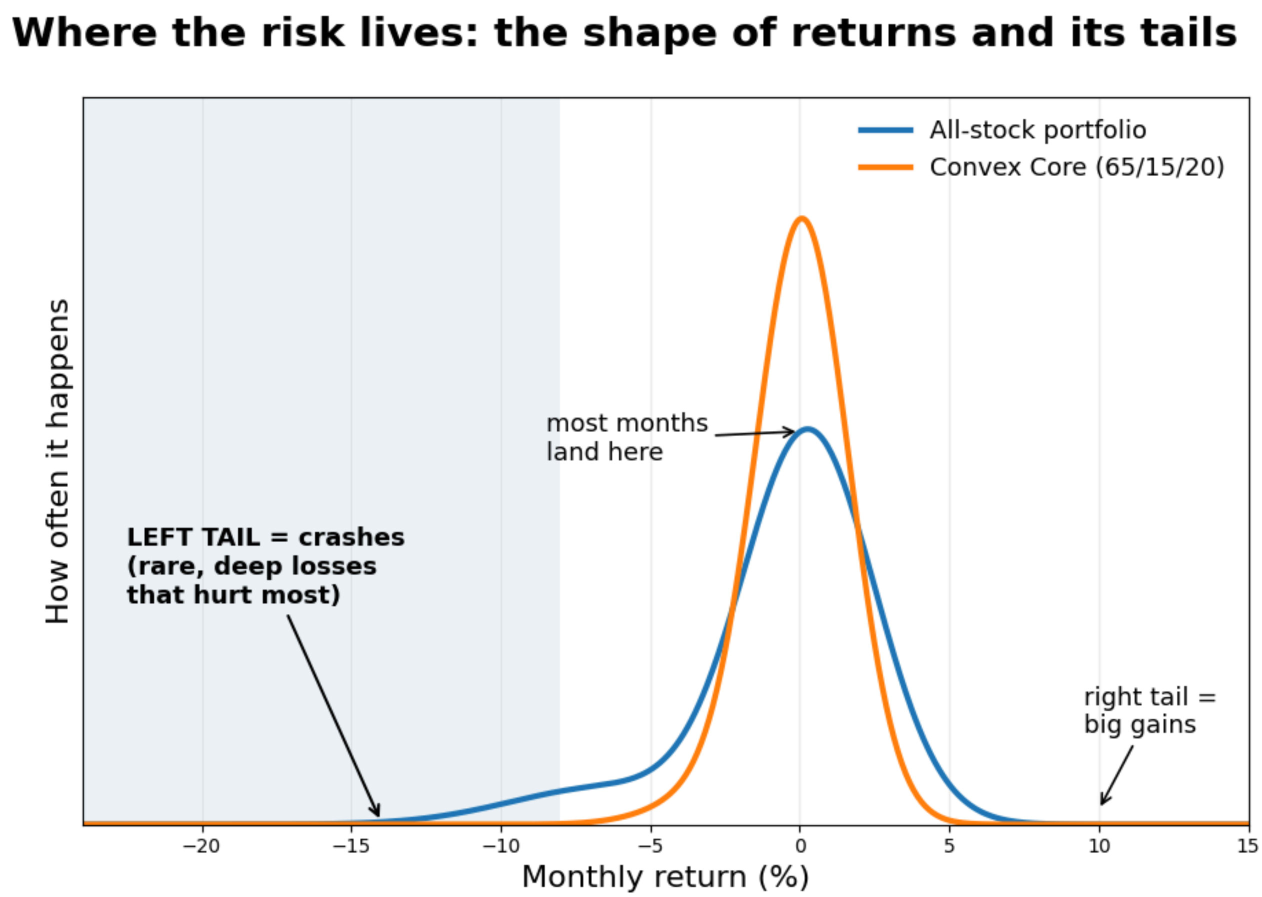


As the stock share climbs from 65% toward 100%, expected return rises gently (green) while the worst-case drawdown deepens sharply (red). The lines cross — you're paying a steep risk price for a small return gain.

Three things happen as you crank equity toward 1.0:

- **Average return rises — modestly.** Going from 65% to 100% stocks lifts expected return by roughly a point or so a year. Real, but small.
- **The holes get much deeper.** A balanced version might lose ~20% in a bad crash; an all-stock version lost about 50% in 2008. Remember the one idea — recovering from −50% takes +100%. That asymmetry quietly eats the extra return.
- **You delete the insurance.** At 95–100% stocks there's almost no duration and no convexity left. The portfolio that sailed through 2022 becomes the portfolio that got hit by it. You've removed the very thing the design exists for.

You can see the cost in the shape of returns. An all-stock portfolio has a fatter, longer **left tail** — the rare, deep losses. The three-sleeve mix pulls that tail in:



Illustrative shapes (not a backtest) to show how the hedges thin the dangerous left tail.

Each curve shows how often different monthly returns happen. The **tails** are the rare extremes at the edges. The left tail — crashes — is what does lasting damage; the hedges make the blue portfolio's left tail thinner and shorter than the all-stock red one.

So "more stocks" isn't wrong — it's a choice to accept deeper holes for a bit more upside. If your horizon is very long, you never withdraw in a downturn, and you can truly ignore a 50% drop, a higher equity weight may suit you. If you'd panic-sell at the bottom, or you'll be drawing income, the deep left tail is exactly the risk the 65/15/20 mix is built to tame.

How it behaves in different crashes

TYPE OF CRASH	RECENT EXAMPLE	WHAT FALLS	WHAT PROTECTS YOU
Recession / deflation panic	2008, 2020	Stocks	DURATION Treasuries rally
Inflation / rate shock	2022	Stocks and bonds	CONVEXITY trend profits
Normal bull market	most years	—	GROWTH does the work; insurance costs a little

Where to hold what — the tax-smart setup

If you have both a taxable brokerage account and an IRA/Roth, treat them as one portfolio: **tax-friendly things go in taxable; tax-heavy things go in the IRA.**

ACCOUNT	PUT HERE	WHY
Taxable brokerage	`VOO/SPY · VXUS · AVUV · DBMF · VTEB/MUB`	Stocks are taxed gently (low turnover, qualified dividends). DBMF's ETF wrapper is tax-friendly. For bonds, **municipal bonds** (VTEB/MUB) pay federally tax-free interest — they replace Treasuries here to dodge the tax on bond income.
IRA / Roth (tax-free)	`AQMNX/AQMIX · IEF/VGIT/TLT`	The AQR fund trades constantly and Treasury interest is taxed as ordinary income — both would be taxed heavily in a brokerage account. Inside an IRA, that turnover and income cost you nothing.

> Honest trade-off: municipal bonds are an imperfect swap for Treasuries. They give tax-free income but don't jump as hard in a true flight-to-safety panic. If deflation-crash protection is the priority, real Treasuries (in the IRA) are the stronger hedge; munis are the tax-friendly compromise for the taxable side.

Measuring volatility — and using it to lower your risk

Here's a simple, repeatable tool that historically improves your risk-adjusted returns (more return per unit of stomach-churn). The idea: you can't predict returns, but volatility is predictable — wild days cluster together. So when the market gets wild, briefly hold fewer stocks; when it calms down, go back to full.

Step 1 — Measure volatility (about 10 minutes a month)

1. Pull the **daily closing prices** of your main stock fund (say VOO) for the last **21 trading days** — roughly one month. Any free site or spreadsheet does this.
2. Turn them into **daily % changes** (today ÷ yesterday − 1).
3. Take the **standard deviation** of those daily changes (one spreadsheet function: =STDEV()).
4. Annualize it: **multiply by 16** (more precisely $\sqrt{252} \approx 15.9$). That number is your annualized volatility — call it your "weather reading."

```
worked example
daily standard deviation  =  1.9% / day
annualized volatility  =  1.9% × 16  ≈  30% / year  ←  that's high (calm markets sit near
12–15%)
```

Step 2 — Use it (the "volatility brake")

Pick a
target volatility — 15% is a reasonable everyday number. Then set how much of your stock sleeve to actually hold:

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stocks to hold  =  min( 100%,   target ÷ your reading )

reading 15%  →  15÷15 = 100%   (hold it all — never use leverage)
reading 30%  →  15÷30 =  50%   (hold half; park the rest in a money-market / T-bill fund)
reading 45%  →  15÷45 ≈ 33%   (hold a third)
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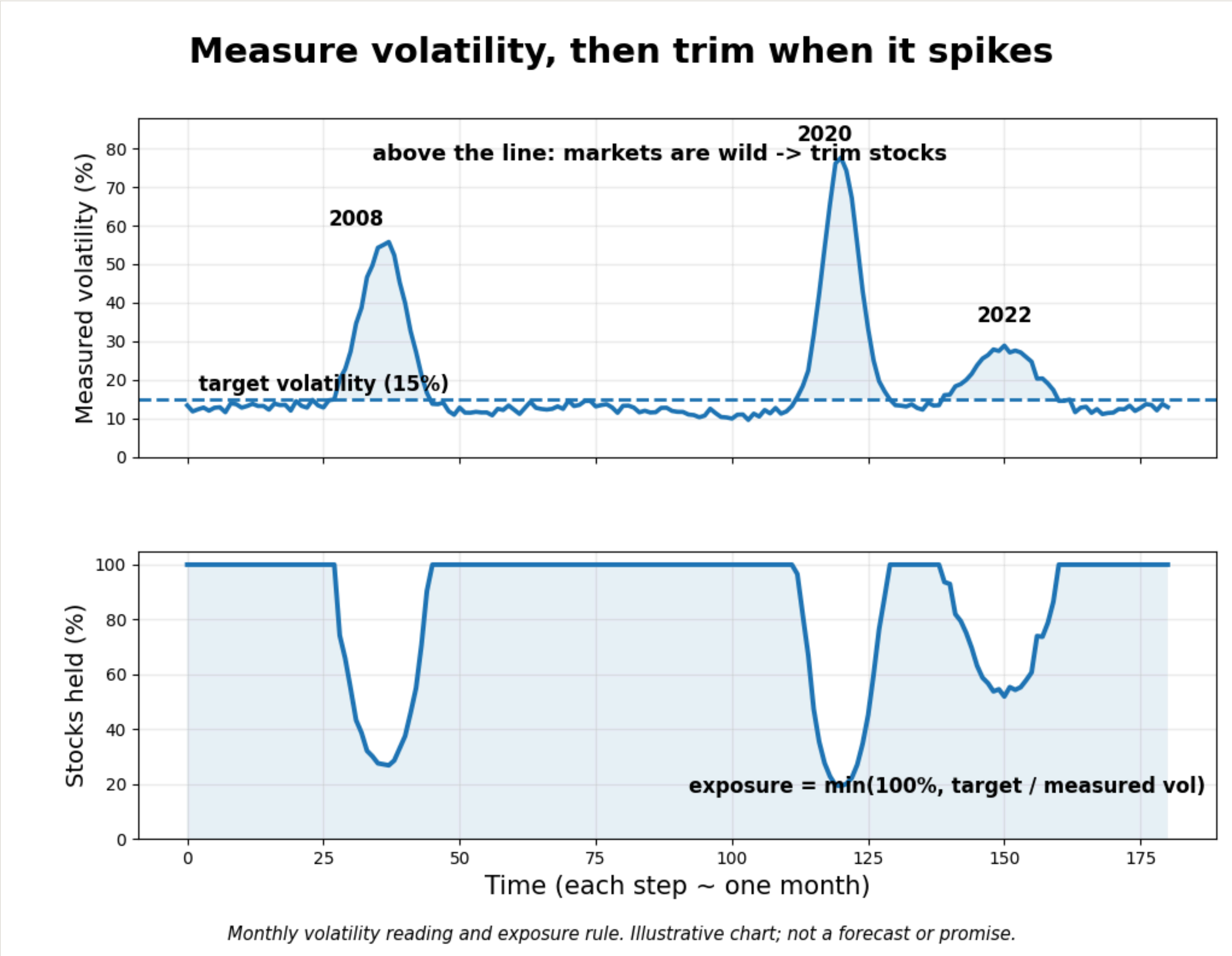
When the storm passes and your reading falls back to 15%, you step back to 100%. That's the whole rule. The chart below shows it running through three real stress periods — volatility spikes, you trim; it settles, you re-enter.



Top: your monthly volatility reading. Whenever it pokes above the 15% target line (shaded), markets are wild. Bottom: the share of stocks you hold automatically drops during those spikes and recovers as things calm — no forecasting, just a thermostat.

Why it helps: the worst market days bunch together, so trimming during high-volatility stretches sidesteps a chunk of the deepest drops — which (back to the one idea) is where compounding is won or lost. Historically this raised return-per-unit-of-risk for stocks.

> Keep it honest and simple: check monthly, not daily, so you don't over-trade. It isn't magic — sometimes it trims right before a sharp V-shaped rebound and you miss some upside. Run it inside an **IRA** where the trimming creates no tax bill. And if this feels like too much, skip it — the three-sleeve structure already does most of the work.

The simple rules for running it

1. **Pick your weights** (e.g., 65 / 15 / 20) and buy the funds.
2. **Leave it alone.** No predicting, no tinkering.
3. **Rebalance once or twice a year** back to target — or steer new contributions into whatever sleeve has lagged. In a crash that means trimming the hedge that paid off and buying stocks while they're cheap.

4. **(Optional) run the volatility brake** from the previous section on your stock sleeve. Powerful, but skippable.

Honest caveats — read before you buy

Trend-following is insurance, and insurance has a cost. In calm, choppy markets DBMF and the AQR fund will lose money for long, boring stretches. That's the premium — don't sell it because it lagged in a bull market.

Managed futures cost more than index ETFs, and the AQR fund is tax-inefficient — hence the IRA placement.

You will trail a roaring bull market. By design, ~35% of this isn't in stocks. The payoff comes in the bad years.

The charts here are illustrative, drawn to teach the ideas — not backtested results, and not a promise. Markets can behave in new ways.